

Regex vs PCRE2 JIT — Benchmark Results

Ratio = PCRE2 JIT ÷ Regex. >1x = Regex faster. JIT compile time excluded.

OS	macOS-26.5.2-arm64-arm-64bit-Mach-O
Python	3.14.6
CPU	Apple M3 Pro
RAM	36 GB
Mojo	Mojo 1.0.0 (ed45d567)

Benchmark	Regex (µs)	PCRE2 JIT (µs)	Ratio
Throughput scaling			
throughput_literal_100B	0.006	0.010	1.8x
throughput_literal_10KB	0.102	0.249	2.4x
throughput_literal_100KB	0.954	2.318	2.4x
throughput_literal_1MB	10.086	23.060	2.3x
throughput_class_10KB	0.301	2.733	9.1x
throughput_nomatch_100KB	0.950	8.275	8.7x
Anchors			
anchor_bol	0.000	0.008	20.0x
anchor_eol	0.006	0.009	1.4x
anchor_word_boundary	0.005	0.011	2.2x
anchor_word_boundary_miss	0.009	0.010	1.1x
anchor_bol_miss_10KB	0.001	0.007	10.0x
Multiline / DOTALL			
dotall_multiline_body	0.002	0.013	5.7x
Named groups			
named_group_date	0.003	0.015	4.4x
named_group_email	0.009	0.019	2.1x
positional_group_email	0.011	0.019	1.8x
Negative lookahead			
neg_lookahead	0.005	0.009	1.7x
neg_lookbehind	0.004	0.009	2.2x
password_validation_lookahead	0.022	0.028	1.3x
Alternation			
alternation_4	0.002	0.009	5.3x
alternation_4_search_2KB	0.193	2.253	11.7x
alternation_16	0.001	0.011	12.2x
alternation_16_miss	0.000	0.025	60.0x
Replace			
replace_50_matches	1.031	1.722	1.7x
replace_named_backref	0.037	0.096	2.6x

Benchmark	Regex (µs)	PCRE2 JIT (µs)	Ratio
Pathological patterns			
pathological_optional_16	0.010	105.020	10398.0x
pathological_dotstar_anchored_5K	0.119	2.150	18.1x
pathological_dotstar_miss_5K	0.001	0.110	100.0x
pathological_triple_backref	0.008	0.014	1.7x
pathological_nested_quantifier_miss	8.273	0.000	0.0x
Other			
static_email_search_2KB	0.109	1.375	12.6x
Real-world patterns			
realworld_url_parse	0.017	0.031	1.9x
realworld_phone	0.005	0.011	2.2x
realworld_hex_color	0.003	0.009	2.9x
realworld_semver	0.013	0.021	1.6x
realworld_ws_normalize	0.102	0.166	1.6x
realworld_log_search_1000_lines	0.803	0.780	1.0x
Inline flags			
inline_ignorecase	0.004	0.010	2.5x
inline_multiline_search	0.006	0.011	1.8x
Engine comparison			
engine_dfa_no_capture	0.004	0.010	2.4x
engine_pike_with_capture	0.007	0.015	2.3x
engine_backtrack_with_backref	0.008	0.012	1.5x
Regex faster: 39 slower: 2			